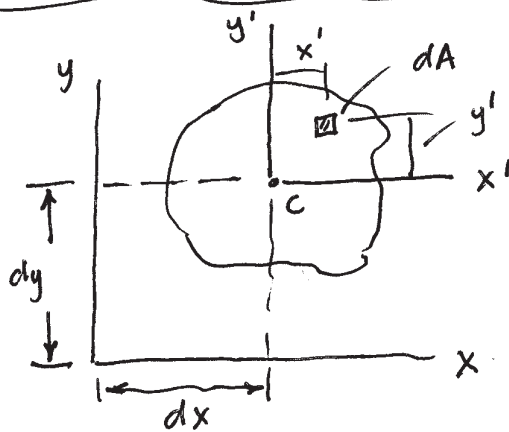


Derive PAT (Parallel Axis Theorem) for POI (I_{xy} , Product of Inertia)



The POI (I_{xy})
for dA is:

About the
x-y
axes...

$$dI_{xy} = (\underbrace{dx + x'}_{\tilde{x}})(\underbrace{dy + y'}_{\tilde{y}}) dA$$

For the entire area:

$$I_{xy} \int_A \tilde{x} \tilde{y} dA = \int_A (dx + x')(dy + y') dA$$

$$= \int_A dx dy dA + \underbrace{\int_A x' y' dA}_{\bar{I}_{x'y'}} + \underbrace{dx \int_A y' dA}_{=0} + \underbrace{dy \int_A x' dA}_{=0}$$

$$dx dy \int_A dA$$

$$= A dx dy$$

$\bar{I}_{x'y'}$
about Area's
centroidal axes

since about
centroidal axis

Finally,
$$I_{xy} = \bar{I}_{xy} + A dx dy$$

which is the PAT for the Product of Inertia